

Vu Thanh Loc (Vu) Mai

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EDUCATION

Queen's University September 2022 – Present
Bachelor of Computing (Honours), GPA: **4.26/4.30** Kingston, ON

- **International Admission Awards** valued at **\$100,000** (May 2022)
- **Dean's Honour List with Distinction** (2023–2024, 2024–2025)

TECHNICAL SKILLS

- **Programming Languages:** C++, Python, Java, Bash, JavaScript, HTML/CSS, SQL, GraphQL
- **Libraries/Frameworks:** React, Flask, Express, Django, Next
- **Developer Tools:** Jira, Git, Docker, Kubernetes, OpenShift, Terraform, GitHub Actions, AWS
- **Practices:** Agile, CI/CD, Unit Testing, SDLC

EXPERIENCE

Research Assistant May 2024 – September 2024
RISE Lab, Queen's School of Computing Kingston, ON

- Conducted advanced analysis of **GitHub Actions** workflows to optimize cache performance by developing **Python** scripts that extract cache-related settings from YAML files in git diff patches.
- Implemented atomic file writes for processing **282** repositories (**60 GB**).
- Achieved high performance by processing each repository (with an average of **1700** commits) in **22 seconds**.
- Published research findings in the [ACM Digital Library](#).

Teaching Assistant September 2024 – Present
Queen's University Kingston, ON

- Assisted in grading **Python** and **C** assignments for Data Structures and Computer Architecture courses.
- Dedicated over **5** hours weekly for office hours and scheduled lab sections to resolve problems for **40+** students on average.

ACTIVITIES

Technical Team Member October 2023 – April 2024
Queen's Google Developer Student Club Kingston, ON

- Collaborated with 5 peers to develop and maintain the club's website. Leveraged **Next.js**, **Tailwind CSS**, and **Chakra UI** to create a dynamic, server-side rendered website.
- Gained **100+** unique visitors in the first month of launching the website ([Website](#)).

Team Lead January 2024 – April 2024
Qweb - Queen's University Web Development Club Kingston, ON

- Led a 5-person team to develop a dynamic quiz platform with sticky sessions using **Redis** and **Node.js**.
- Streamlined Agile workflows with **Jira**, reducing delivery time by **20%**.

PROJECTS

- 🔗 **SlideFlow - Presentation AI Controller (QHacks 2025 Winner)** January 2025
 - Developed an AI-powered PDF tool enabling voice-controlled search via the **Webspeech API**.
 - Integrated **Google Cloud Vision API** and **Redis** cache; utilized **S-BERT** for semantic evaluation.
 - Built a seamless front end with **React** and **Tailwind CSS**, connected to a **Flask** back end.
 - Won QHacks' 2025 "Best Use of AI" award, finishing in the top **4** of 80 teams.
- 🔗 **Flight Booking Platform** September 2024 – December 2024
 - Developed a 3-tier application with **React** (front-end), **Flask** (back-end), and **MongoDB** (database).
 - Achieved **92%** backend test coverage using **pytest**; conducted unit testing for UI components with **React Testing Library** and **Jest**.
 - Performed integration and end-to-end testing with **Cypress**.
- 🔗 **Elastic Cloud Storage** November 2024 – December 2024
 - Developed a cloud storage service using a serverless architecture with **AWS S3**, **CloudFront**, **API Gateway**, **Lambda**, and **DynamoDB**.
 - Enabled automatic deployment to AWS with **Terraform** and implemented a robust **GitHub Actions** CI/CD pipeline.